QTW – 7333 – 2/16/2025
As usual summariy and study guide found here:
https://rpubs.com/Texaschikkita/Mod7_DecisionTrees
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Presession Module 7

Relevant Questions from asynch:

1. **Entropy vs Gini Confusion:**
   When building decision trees in my coding practice, I keep second-guessing: *Should I default to entropy or Gini?* In real-world scenarios, do you find data characteristics (like class imbalance) make one metric clearly better than the other? I've read they often give similar results, but I'm curious about edge cases where they might meaningfully differ.
2. **The Goldilocks Zone for Random Forests:**
   Our textbook mentions using "enough trees" in random forests - but what's *enough*? Between 100 trees that take 10 minutes to train vs 500 that take an hour, how do you decide what's worth waiting for computational-y? Any rules of thumb you use in your own research?
3. **Overfitting Endgame Strategies:**
   I've experimented with max depth constraints and minimum samples per leaf, but these often feel like blunt instruments. How do *you* approach pruning vs pre-growth constraints? Are there advanced techniques (like cost-complexity pruning) that students should prioritize learning?

Takeaways (What clicked for me this week in asynch)

1. **Decision Trees Are Information Hunters:**
   Realized both entropy and Gini impurity are essentially yardsticks for measuring how much clarification we get from each split - lower disorder = better split. This makes feature selection feel more purposeful!
2. **Bagging = Teamwork for Models:**
   The bagging concept finally makes sense - like assembling a council of slightly different decision trees to vote, where each member gets their own "opinion" (subsampled data). Reduces that awkwardness when one tree overreacts to noise.
3. **Random Forest's Secret Sauce:**
   Aha moment: The "random" in random forests does double duty - random data

samples *and* random feature subsets. This dual randomness acts like noise-cancelling headphones against overfitting, letting the true signal shine through.